

Hacettepe University - Computer Engineering CMP720 Embedded System Design - Spring 2026

Final Project Topic Proposal Template

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Project Topic Title

Quadrotor Control and Evaluation Simulation on NVIDIA Jetson

Project Track

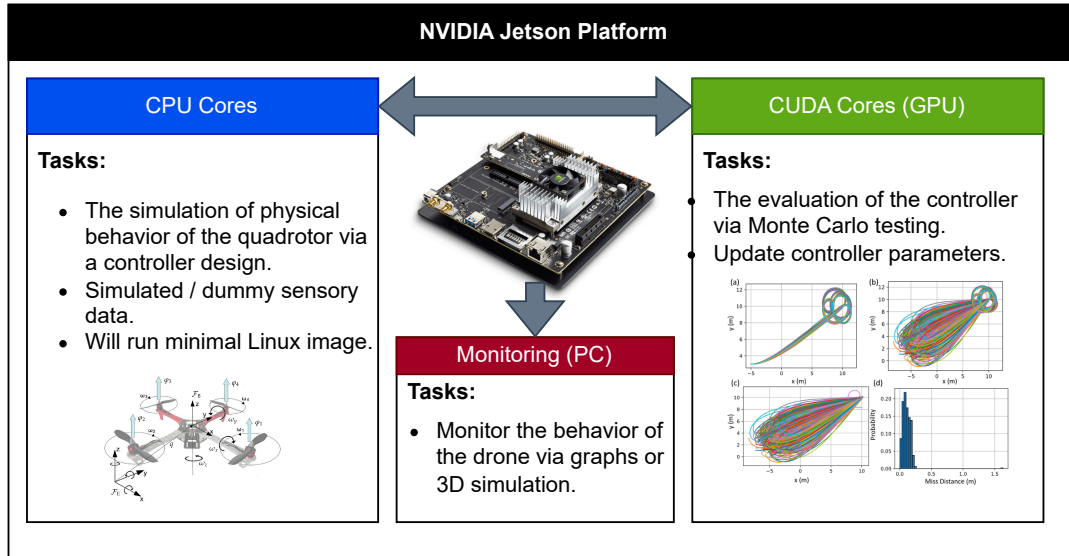
- Implementation Track

Abstract

Unmanned aerial vehicles (UAV's) rely heavily on control algorithms to maintain their stability and execution of their operation. Drone controllers must operate reliably under many different conditions such as sensor noise, external disturbances, model uncertainties, and varying initial states. Designing a robust controller considering all uncertainties is an extremely difficult task. One common way to evaluate robustness is through Monte Carlo simulation, where the same controller is tested many times under randomized scenarios. However, this approach requires running a very large number of simulations, which can become computationally expensive when performed only on a CPU. This makes controller evaluation slower and limits the number of test cases that can be analyzed. Also since the needs can vary on the actual flight the controller designed on the ground may fail during the operation, since it does not have the tool for updating its control algorithm.

In this project, we aim to address these limitations by using GPU acceleration for Monte Carlo-based drone controller evaluation and in addition, we plan to implement the approach on an embedded CPU + GPU platform, such as an NVIDIA Jetson board, so that the system can be tested in a more realistic embedded computing environment. The idea is that the control of the quadrotor according to the designed controller will run on CPU as a simulation of the drone while the controller evaluation and update part will run on GPU.

The expected outcome is a proof-of-concept embedded prototype showing that a drone simulation can keep its nominal control loop on the CPU while using the GPU to evaluate many possible near-future outcomes in parallel. This would demonstrate a practical use of embedded GPU resources beyond graphics or perception tasks, provide quantitative CPU-versus-GPU comparisons, and create a reusable structure for future work on scenario-based controller evaluation, controller adaptation, and sim-to-real oriented experimentation.



Motivation / Societal Impact

There is a need for adaptable controller algorithms in robotics field and many proposed ways to make them. Mostly they are being tested and evaluated on desktop hardware however, drones operate under very different constraints and control algorithms on such devices should be adaptable to the changing environment. To solve the adaptable controller problem we think that it is desirable to have an lightweight hardware to calculate and update the controller. This is why the NVIDIA Jetson boards are chosen as the candidate platform since on A board such as NVIDIA Jetson which offer both CPU cores and a CUDA-capable GPU within a single board, making it a suitable platform for testing whether a realistic division of labor between the two processors can support advanced decision-support computations.

The proposed embedded drone simulation platform offers several advantages. It allows early validation of control algorithms in an environment that closely resembles the final deployment hardware, reduces development costs associated with physical testing, and provides a portable testbed for rapid experimentation. Additionally, the system can serve as a hardware-in-the-loop (HIL) or software-in-the-loop (SIL) environment, facilitating incremental development and verification of autonomous flight capabilities. Also, if the proposed architecture comes out efficient it may have a place among many AI driven sim2real algorithms regarding control of the robotic systems. Overall, implementing a drone control simulation system on an embedded platform bridges the gap between algorithm development and real-world deployment, enabling more reliable, efficient, and scalable UAV control system development.

Related Work / References

Quadrotor Real-Time Simulation: A Temporary Computational Complexity-Based Approach [1].

CUDA for Rapid Controller Robustness Evaluation: A Tutorial [2].

Attitude Control System for Quadrotor Using Robust Monte Carlo Model Predictive Control [3].

References

- [1] G. Delgado-Reyes, J. S. Valdez-Martínez, M. Hernández-Pérez, K. R. Pérez-Daniel, and P. J. García-Ramírez, “Quadrotor real-time simulation: A temporary computational complexity-based approach,” *Mathematics*, vol. 10, no. 12, p. 2032, 2022.
- [2] R. Melzer, M. Gandhi, R. Schlossman, K. Williams, and J. Parish, “Cuda for rapid controller robustness evaluation: A tutorial,” tech. rep., Sandia National Laboratories, Sept. 2021. Accessed: 2026-03-08.
- [3] K. Masuda and K. Uchiyama, “Attitude control system for quadrotor using robust monte carlo model predictive control,” *Actuators*, vol. 13, no. 11, p. 443, 2024.